


CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

Test Pattern

JEE(Advanced)
FULL SYLLABUS
 24-04-2024

JEE(Main + Advanced) : ENTHUSIAST COURSE ALL STAR BATCH (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY
GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **20** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

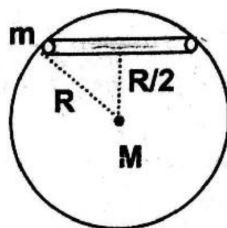
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Consider the earth as a uniform sphere of mass m and radius R . Imagine a straight smooth tunnel made through the earth which connects any two points on its surface as shown in the figure. The particle is released from rest as shown in the figure. Find the time when it comes at the same point ($g = GM / R^2$)



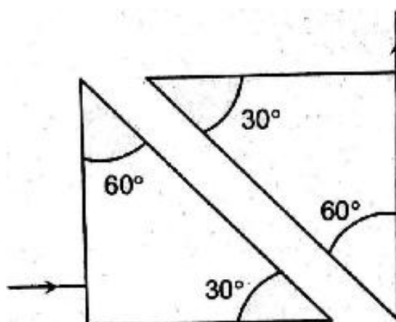
(A) $2\pi\sqrt{\frac{R_e}{g}}$

(B) $2\pi\sqrt{\frac{3R_e}{4g}}$

(C) $2\pi\sqrt{\frac{R_e}{2g}}$

(D) $2\pi\sqrt{\frac{3R_e}{g}}$

2. Two right angled prisms having the same refracting angle 60° are placed as shown in the figure. A ray of light incident on the first prism is finally deviated by an angles 90° grazingly as shown. The refractive index of second prism is $\sqrt{2}$. The refractive index of the first prism is equal to



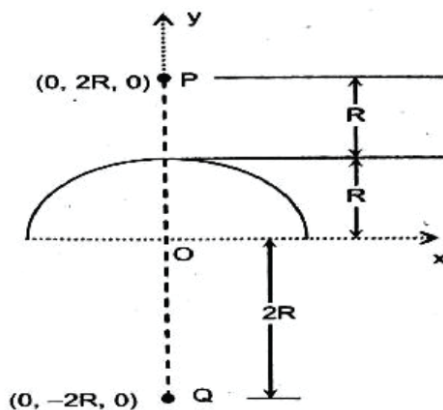
(A) $\sqrt{2} \frac{\sin 30^\circ}{\sin 60^\circ}$

(B) $\sqrt{2} \frac{\sin 45^\circ}{\sin 60^\circ}$

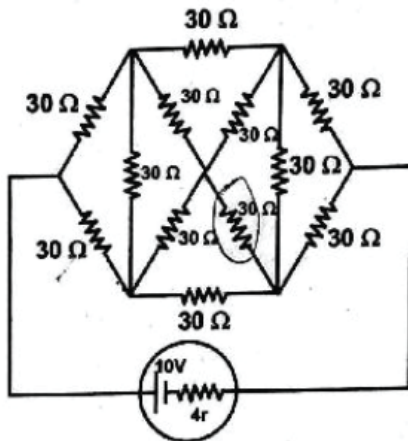
(C) $\sqrt{2} \frac{\sin 15^\circ}{\sin 60^\circ}$

(D) $\sqrt{2} \frac{\sin 60^\circ}{\sin 30^\circ}$

3. A hemisphere of radius R and density ρ is held in space as shown in figure. If gravitational field intensity at P is $E \hat{(-j)}$ then magnitude of gravitational field intensity at Q is



- (A) $\left(E - \frac{\pi GR\rho}{3}\right) \hat{j}$
 (B) $\frac{\pi GR\rho}{3} \hat{(-j)}$
 (C) $\left(E - \frac{\pi GR\rho}{3}\right) \hat{(-j)}$
 (D) $\left(E + \frac{\pi GR\rho}{3}\right) \hat{j}$
4. Find the value of r so that maximum power is generated in external circuit.



- (A) $r = 10\Omega$
 (B) $r = 20\Omega$
 (C) $r = 30\Omega$
 (D) $r = 40\Omega$

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

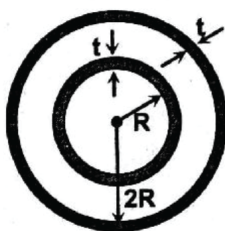
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

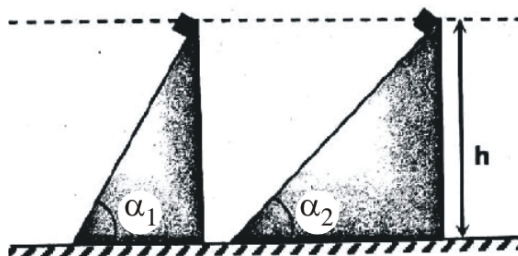
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. A spherical shell of volume mass density ρ , thickness t and radius R ($R \gg t$) is placed concentrically inside another shell of radius $2R$ having same thickness and of same material as shown in figure. The circumferential stress developed in the outer shell is

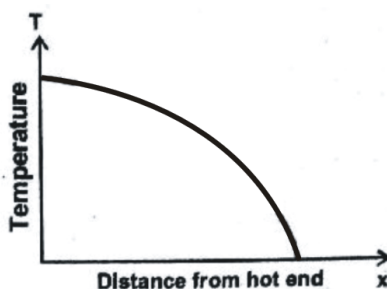


- (A) Compressive in nature (B) Tensile in nature
- (C) $3G\pi R t \rho^2$ (D) $\frac{3}{2}G\pi R t \rho^2$
6. An ${}_8\text{O}^{16}$ nucleus is spherical having radius R , surface area A and volume V . According to empirical observations, the volume of the ${}_{50}\text{Sn}^{128}$ assumed to be spherical is V' , radius R' and surface area A' . Then
- (A) $R' = 8R$ (B) $A' = 8A$
- (C) $V' = 8V$ (D) $R' = 2R$

7. Two objects of the same mass slide down on two different smooth fixed inclined plane of same height but of different angles. When they reach the bottom, then I_1 and I_2 are the impulses of the weight of the first and the second object during the motion, P_1 and P_2 are the powers developed by the weight of the first and second objects.

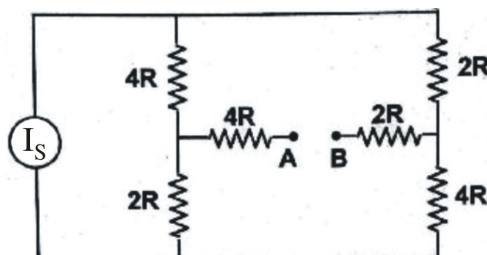


- (A) $\frac{I_1}{I_2} = \frac{\sin \alpha_2}{\sin \alpha_1}$ (B) $\frac{I_1}{I_2} = 1$
 (C) $\frac{I_1}{I_2} = \frac{\sin \alpha_1}{\sin \alpha_2}$ (D) $\frac{P_1}{P_2} = \frac{\sin \alpha_1}{\sin \alpha_2}$
8. The ends of a long bar are maintained at different temperature and there is no loss of heat from the sides of the bar due to conduction or radiation. The graph of temperature against distance from the hot end of the bar at steady state is shown in the diagram. From this we can conclude that



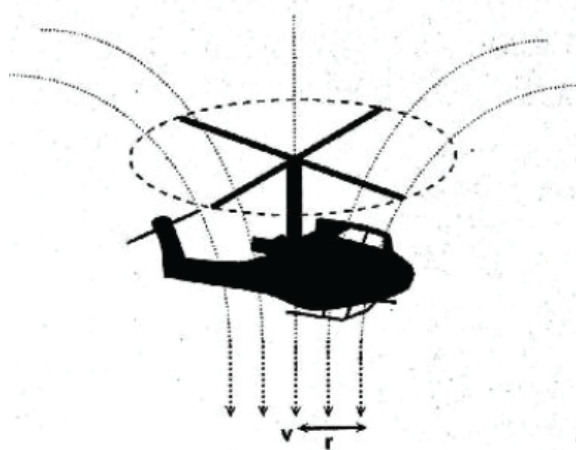
- (A) The cross-sectional area of the bar may be increasing as that distance from the hot end increases
 (B) The cross-sectional area of the bar may be decreasing as the distance from the hot end increases
 (C) The thermal conductivity of the bar may be decreasing as the distance from the hot end increases
 (D) The thermal conductivity of the bar may be increasing as that distance from the hot end increase

9. Consider the circuit shown below. I_S is constant current source, meaning that no matter what device is connected between points A and B, the current provided by the constant current source is always (I_S) the same

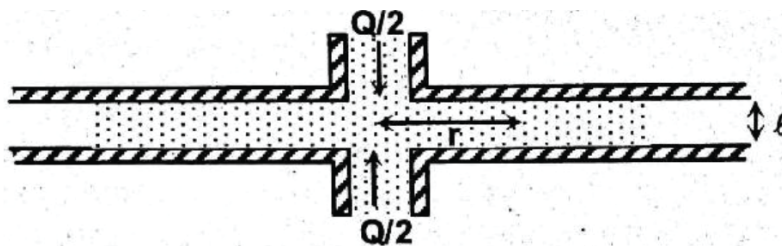


- (A) If an ideal voltmeter is connected between A and B, the reading of voltmeter is $I_S R$.
- (B) If an ideal voltmeter is connected between A and B, the reading of voltmeter is $2I_S R$.
- (C) If an ideal ammeter is connected between A and B, the reading of ammeter is $\frac{I_S}{9}$.
- (D) If an ideal ammeter is connected between A and B, the reading of ammeter is $\frac{I_S}{6}$.
10. We have two equal size boxes A and B. Each box contains gas that behaves as an ideal gas. Thermometer are inserted into each box and it is found that the gas in box A is at a temperature of 50°C , while the gas in box B is at 10°C . Which of the following statements(s) is/are true?
- (A) The pressure in A may be higher than in B.
- (B) There may be more molecules in A and than in B.
- (C) The molecules in A have more average kinetic energy per molecule than those in B.
- (D) The molecules in A are moving faster than B.

11. A helicopter has a mass m and maintains height by imparting a downward momentum to a column of air defined by the slipstream boundary as shown in figure. The propeller blades can project a downward air speed v , where the pressure in the stream below the blade is atmospheric and the radius of the circular cross section of the slip stream is r . Neglect any rotational energy of the air, the temperature rise due to air friction and any change in air density (ρ)



- (A) $V = \frac{1}{r} \sqrt{\frac{mg}{\pi \rho}}$ for equilibrium
- (B) $v = \frac{2}{r} \sqrt{\frac{mg}{\pi \rho}}$ for equilibrium
- (C) Power of engine = $\frac{mg}{2r} \sqrt{\frac{mg}{\pi \rho}}$ for equilibrium
- (D) Power of engine = $\frac{mg}{r} \sqrt{\frac{mg}{\pi \rho}}$
12. Incompressible liquid of negligible viscosity is pumped at total volume flow rate Q through two small holes into the narrow gap between closely spaced parallel plates as shown. The liquid flowing away from the holes has only radial motion. Assume uniform flow across any vertical section.



- (A) Flow velocity is proportional to $\frac{1}{r^2}$
- (B) Flow velocity is proportional to $\frac{1}{r}$
- (C) Pressure difference between the centre and any radial point is proportional to $\frac{1}{r^2}$
- (D) Pressure difference between the centre and any radial point is zero

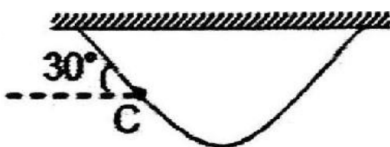
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
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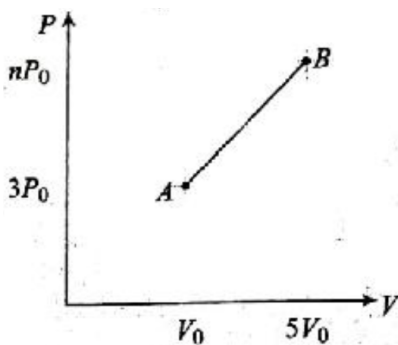
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A uniform rod of length ℓ cross-sectional area A and Young's modulus Y is subjected to two forces ' F ' and ' $2F$ ' at its two ends stretching the rod. The heat generated in the rod is found to be $\frac{(10a + b) F^2 \ell}{12AY}$. Find $a+b$. { a, b : Positive integers}
2. A rope is hanging vertically with its two ends fixed to walls at same horizontal level (as shown). Tension at the bottom most point is 90 N. If the mass/unit length of rope is 4kg/m , find radius of curvature (in metres) at point 'C' {Take $g = 10 \text{ m/s}^2$ }



3. A source of frequency 1800 Hz is moving on a line, with speed 30 m/s. An observer is standing at a point far distant from the line. The wavelength of sound received by the observer when the source is nearest to the observer is $n/11$ metres. Find ' n ' (speed of sound in air = 330 m/s)
4. Rain drops of some initial mass start falling from rest. In their way they gain mass from clouds with a rate proportional to the product of instantaneous mass and instantaneous velocity with proportionality constant being 2.5 SI units. Find the terminal speed attained by drops (in m/s). (Take $g = 10 \text{ m/s}^2$)
5. A sinusoidal wave of wavelength ' λ ', amplitude ' A ' and angular frequency ' ω ' travel on a string of mass/unit length ' μ '. The energy of wave in one wavelength is equal to $\frac{\mu \omega^2 A^2 \lambda}{b}$. Find ' b '
6. One mole of a monatomic ideal gas follows a process $A \rightarrow B$, as shown. The specific heat of the process is $\frac{13R}{6}$. The value of ' n ' on P-axis (see following figure) is



PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

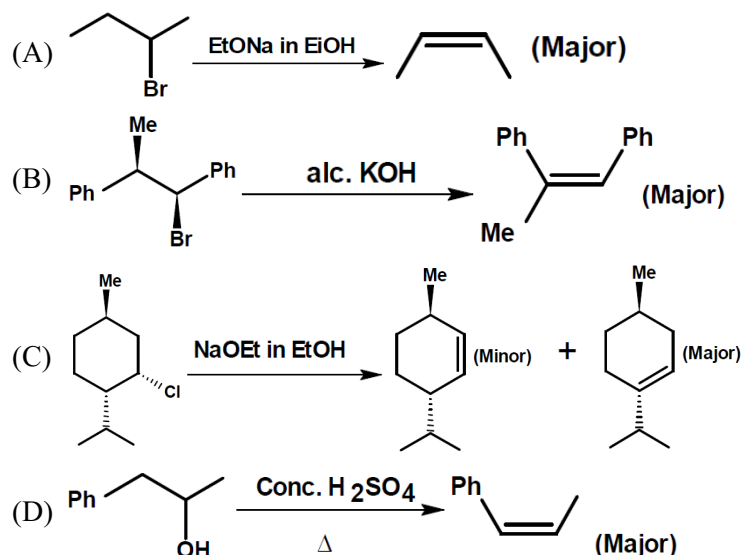
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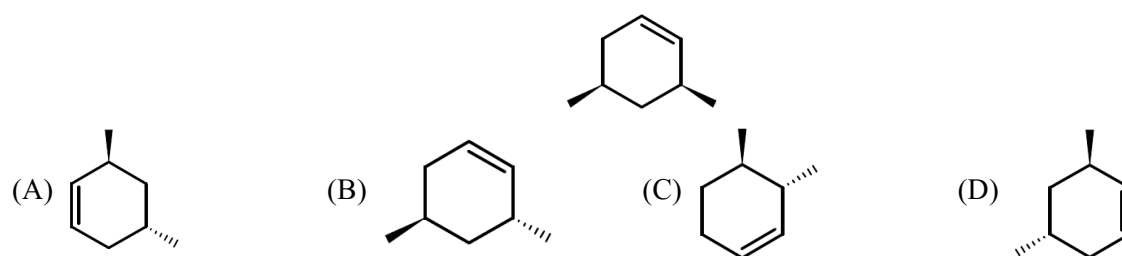
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In which one of the following reactions, products given is correct



2. The structural isomer of given structure is



3. What is the change in Gibbs free energy when 2 moles of an ideal gas at 273K and 2 bar expanded irreversibly to 1 bar
- (A) -3147J (B) +3147J (C) -2270J (D) +2270J
4. To an alkaline salt solution KCN added. Now the solution acidified and FeCl₃ added, then red colouration observed. The salt may posses
- (A) SO₃²⁻ (B) S₂O₃²⁻ (C) NO₂⁻ (D) SO₄²⁻

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
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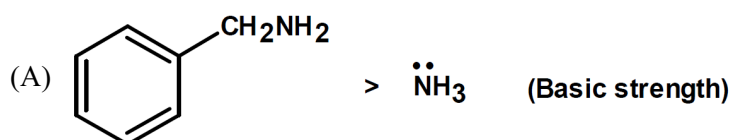
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Which of the following is correct ?



(B) $\text{Et}_2\text{NH} > \text{Me}_2\text{NH}$ (Basic Strength)

(C) $\text{Et}_3\text{N} > \text{Me}_3\text{N}$ (Basic Strength)

(D) $\text{EtNH}_2 > \text{MeNH}_2$ (Basic Strength)

6. Which of the complexes have zero dipole moment, zero magnetic moment and non-zero crystal field stabilization energy (cfse)?

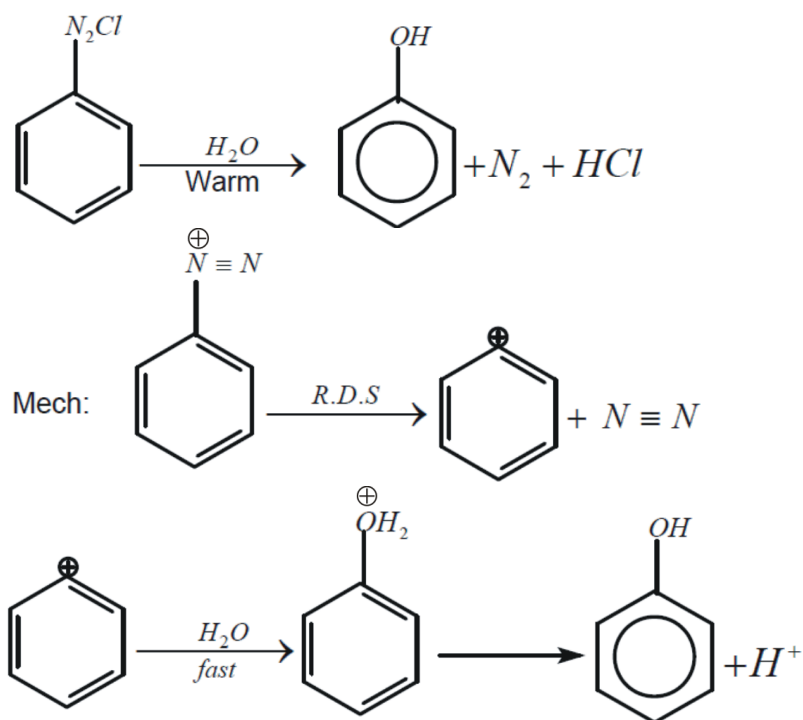
(A) $\text{trans} - [\text{Ni}(\text{en})_2\text{Cl}_2]$

(B) $\text{trans} - [\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$

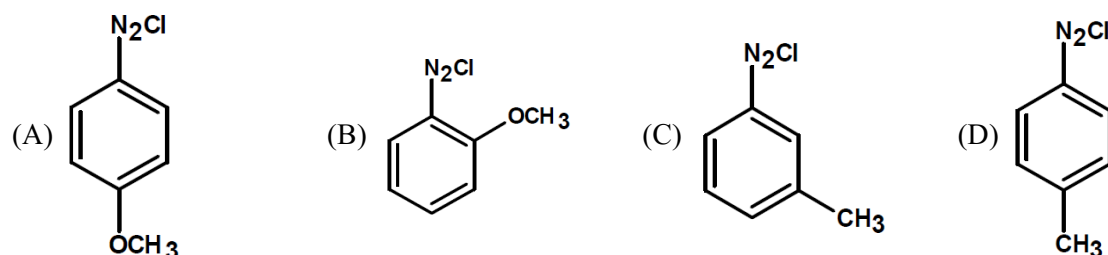
(C) $\text{trans} - [\text{Fe}(\text{CN})_4(\text{H}_2\text{O})_2]^{2-}$

(D) $\text{trans} - [\text{Mn}(\text{H}_2\text{O})_2(\text{Cl})_4]^{2-}$

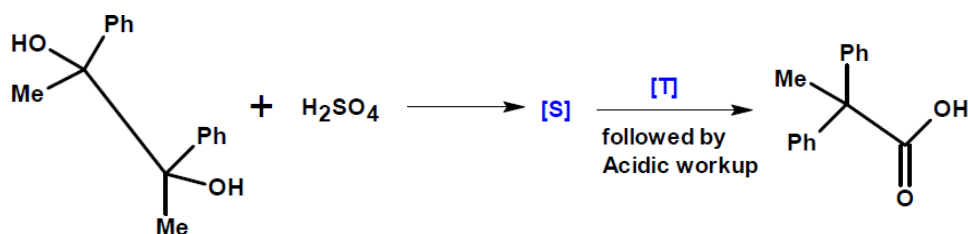
7.



In the above reaction which of the following is/are more reactive than benzene diazonium chloride



8.



(Given, migratory order Aryl > Arkyl > H)

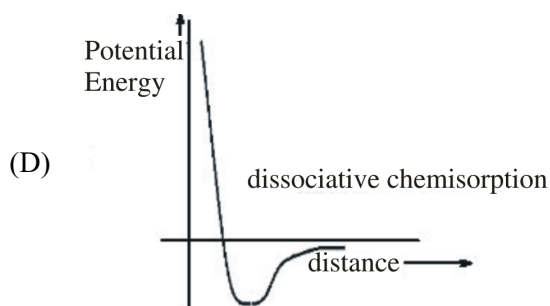
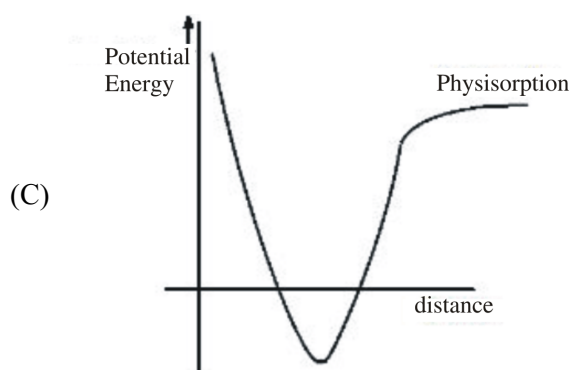
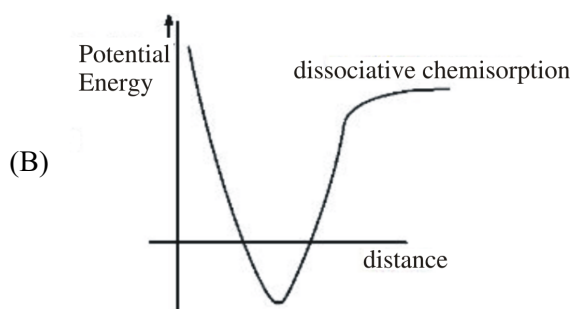
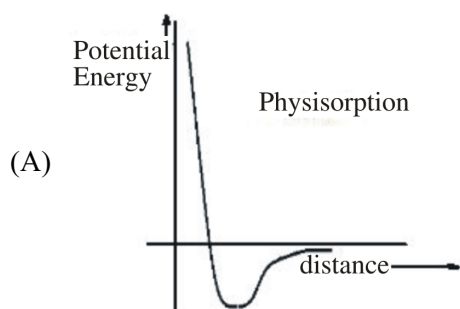
Find, the correct statement(s)

- (A) [T] can be $[\text{Ag}(\text{NH}_3)_2] \text{OH}$
 (B) [T] can be fehling solution
 (C) [T] can be $\text{Cl}_2 + \text{NaOH}$
 (D) [T] can be Benedict solution

9. During the hydrolysis of P_4O_{10} , which of the following acids formed as intermediate/ final product

- (A) Tetrametaphosphoric acid
 (B) Tetrapolyphosphoric acid
 (C) Pyrophosphoric acid
 (D) Phosphoric acid

10. When Z – axis is inter nuclear axis which of the following linear combinations are allowed (i.e. give B.M.O & A.B.M.O)
- (A) $s - p_z$
- (B) $d_{xz} - p_x$
- (C) $d_{x^2-y^2} - d_{x^2-y^2}$
- (D) $d_{yz} - p_z$
11. Consider an element A crystallizing in simple cubic, face centered cubic and body centered cubic lattice. If the radius of an atom of A is r then which of the following statements is / are correct? (assume no change of radius during formation of different crystals)
- (A) Shortest distance between two non – touching spheres in simple cubic unit cell is $\sqrt{8}r$
- (B) Shortest distance between two non – touching spheres in FCC unit cell and SC unit cell is same
- (C) Shortest distance between two non – touching spheres in BCC unit cell is less than that in FCC unit cell
- (D) Void space in BCC unit cell is greater than that in FCC unit cell
12. Which of the following graphs regarding adsorption are correct?



SECTION-II : (Maximum Marks: 18)

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- Answer to each question will be evaluated according to the following marking scheme:

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1. How many acetals are formed when acetaldehyde is treated with $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2(\text{OH})$ in gaseous HCl?
2. Two liquids A and B form a non – ideal solution which obey the equation $P_T = P_A^0 + 3 (P_B^0 - P_A^0) x_B + 2 (P_A^0 - P_B^0) x_B^2$. When equi molar mixture of A and B distilled find the mole fraction of B, when this mixture will have a single boiling point.
3. A 2.0 L flask, initially containing one mole of each CO and H_2O , was sealed and heated to 700 K, where the following equilibrium was established $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$; $K_C = 9$. Now the flask was connected to another flask containing some pure $\text{CO}_2(\text{g})$ at same temperature and pressure, by means of a narrow tube of negligible volume. When the equilibrium was restored, moles of CO was found to be double of its mole at first equilibrium. Determine volume of $\text{CO}_2(\text{g})$ flask in litres
4. A sample of H atoms containing all the atoms is some excited state is irradiated with the light of wavelength λ and atoms get re – excited to further higher excited state. On removing the light source, atoms de – excite to the ground state with the emission of radiations. The emission spectrum thus obtained consists of ten spectral lines. Out of these ten spectral lines, seven lines have wavelengths smaller and only two line have wavelengths longer than λ . The orbit number for initial excited state is
5. The oxidation number of Cr in the product of alkaline oxidative fusion of FeCr_2O_4 is
6. The number of lone pairs of ‘Xe’ present in equatorial position in $[\text{XeF}_5]^-$

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $x \sin A + y \sin B + z \sin C = 0$ and $x^2 \sin 2A + y^2 \sin 2B + z^2 \sin 2C = 0$ then the value of $x^3 \sin 3A + y^3 \sin 3B + z^3 \sin 3C$ is equal to (where $A, B, C; x, y, z \in \mathbb{R}$ and $A + B + C = \pi$)
 (A) 0 (B) Depend on A, B, C
 (C) Depend on x, y, z (D) 3
2. If $g(x)$ is non-negative continuous function such that $g(x) + g\left(x + \frac{1}{2}\right) = 2, \forall x \in \mathbb{R}$, then the value of $\int_0^1 (\sin 4\pi x) g(x) dx$ is
 (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{3}{2}$
3. If there exists atleast one z satisfying $|z - ki| = k + 4$ and $|z - 3| \leq 4, (k \in \mathbb{R})$ and $A = \{0, 4, 8, 10\}$ then how many values k will take from the set A ?
 (A) 0 (B) 1
 (C) 2 (D) 4
4. Let $\vec{p}, \vec{q}, \vec{r}$ are three vectors such that $\vec{p} \times \vec{r} = \vec{q} + \lambda \vec{p}, \vec{p} \cdot \vec{r} = 1$ then which of the following is true
 (A) $2\lambda = \frac{-\vec{p} \cdot \vec{q}}{|\vec{p}|^2}$
 (B) $\vec{r} = \frac{\vec{p} - (\vec{p} \times \vec{q})}{|\vec{p}|^2}$
 (C) $[\vec{p}\vec{q}\vec{r}] = \frac{-2|\vec{p} \times \vec{q}|^2}{|\vec{p}|}$
 (D) $\vec{p} \times \vec{q} = \vec{0}$

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. If P be a point on ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where foci S, S' and tangent at P is at a distance d from origin then $SP \cdot S'P \cdot d^2$
- (A) Independent of 'a' and 'b'
- (B) Independent of ratio $\left(\frac{a}{b}\right)$
- (C) Independent of eccentric angle of point
- (D) Less than or equal to $\max(a^4, b^4)$
6. Consider the planes $P_1 : 2x + y + z + 4 = 0$, $P_2 : y - z + 4 = 0$ and $P_3 : 3x + 2y + z + 8 = 0$. Let L_1, L_2, L_3 be the lines of intersection of the planes P_2 and P_3 , P_3 and P_1 , and P_1 and P_2 respectively. Then which of the following is/are true?
- (A) Atleast two of the lines L_1, L_2 and L_3 are non-parallel
- (B) Atleast two of the lines L_1, L_2 and L_3 are parallel
- (C) The three planes intersect in a line
- (D) The three planes form a triangular prism.

7. $f(x)$ is the function such that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ if $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$ then
- (A) $a = \frac{-5}{2}$ (B) $b = \frac{-3}{2}$
 (C) $a = \frac{-7}{2}$ (D) $b = \frac{-5}{2}$
8. A die is rolled three times. Let E_1 denote the event of getting a number larger than the previous number each time and E_2 denote the event that the numbers (in order) form an increasing A.P, then
- (A) $P(E_2) \leq P(E_1)$ (B) $P(E_1 \cap E_2) = \frac{1}{36}$
 (C) $P(E_1) = \frac{10}{3}P(E_2)$ (D) $P(E_1) = \frac{1}{3}$
9. If the curve $f(x) = \int x^{27} (1 + x + x^2)^6 (6x^2 + 5x + 4) dx$ passes through $(0,0)$, then
- (A) $f(1) = \frac{3^7}{7}$ (B) $f(1) = \frac{2^7}{7}$
 (C) $f(-1) = \frac{1}{7}$ (D) $f(-1) = \frac{3^7}{14}$
10. Normals drawn at points P, Q and R lying on the parabola $y^2 = 4x$ intersect at $(3,0)$ then
- (A) Centroid of $\Delta PQR = \left(\frac{3}{2}, 0\right)$
 (B) Circum centre of $\Delta PQR = \left(\frac{5}{3}, 0\right)$
 (C) Area of $\Delta PQR = 2$
 (D) Radius of circum circle of ΔPQR is $\frac{5}{2}$
11. Let $f(x)$ be a cubic polynomial which is having local maximum at $(1,2)$ and $f'(x)$ has local extrema at $x = 0$. If $f(0) = 1$ then
- (A) Number of real root of $f(x) - f(-x) = 6x - 10$ is 1
 (B) If $g(x) + f(x) = 0$ then complete interval in which $g(x)$ is increasing is $x \in \mathbb{R} - (-1, 1)$
 (C) The range of values of k for which $f(x) = k$ has three distinct solutions is $k \in (0, 2)$
 (D) Range of the function $P(\theta) = f(\sin \theta)$ where $\theta \in \mathbb{R}$ is $[0, 2]$
12. If $\sin^2 x - a \sin x + b = 0$ has only one solution in $(0, \pi)$ then which of the following statements are correct?
- (A) $a \in (-\infty, 1] \cup [2, \infty)$ (B) $b \in (-\infty, 0] \cup [1, \infty)$
 (C) $a = 1 + b$ (D) $a = b$

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Let $\triangle ABC$ right angle at A is inscribed in hyperbola $xy = 4$. If slope of BC is 2 and coordinates of A are (α, β) , then the value of $\frac{\alpha^2}{3\beta^2}$ is equal to
2. A dice is randomly rolled five times the probability that outcome of fifth throw is atleast one of the outcomes of first 4 throws is $\frac{pqr}{6^4}$, then the value of $p + q + r$ is (where pqr is a three digit number)
3. Let A,B be non-singular matrices of order 3 such that $\det(A) = 5$ and $A^{-1}B^2 + AB = 0$, then the value of $\det(A^6 - 2A^4B + A^2B^2)$ is equal to 10^k where k is equal to
4. The area of the quadrilateral, whose vertices are the root of the equation $1 + z - z^3 - z^4 = 0$, is
5. Let $L_1 : 3x + 4y - 1 = 0$ and $L_2 : 5x - 12y + 2 = 0$ be given two lines let image of every point on L_1 w.r.t line L lies on L_2 then one of the possible equation of L can be $ax + by - 23 = 0$, $a, b \in I$, where $\frac{b}{a}$ is equal to
6. Let AB be any chord of the circle $x^2 + y^2 - 2x - 6y - 6 = 0$ which subtends right angle at the point (2,4) then the locus of the midpoint of AB is $x^2 + y^2 + ax + by + c = 0$ where $|a + b + c|$ is equal to

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NAME OF THE CANDIDATE

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I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

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